

EXHIBIT 36

EXHIBIT 36

U.S. Patent No. 9,865,546 Infringement Chart

Title: Contacts to semiconductor substrate and methods of forming same

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Abstract: An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

Claim	Tower '546 Accused Process
[1pre] A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:	To the extent the preamble is limiting, Tower's RF processes (including its 65nm RF process) (hereinafter, "Tower '546 Accused Process") comprises a method for forming a contact in a dielectric layer to a semiconductor substrate, as explained further in the limitations below.¹ For example, the Tower '546 Accused Process is used to manufacture the Qorvo QM81026 chip. The Tower '546 Accused Process comprises forming a contact in a dielectric layer to a semiconductor substrate, as shown in the reverse engineering analysis below, which shows a cross-section of the Qorvo QM81026 chip.

¹ GlobalFoundries reserves the right to identify to identify different or additional Tower '546 Accused Processes and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

EXHIBIT 36

Claim	Tower '546 Accused Process
	
[1a] forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate	The Tower '546 Accused Process comprises forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate. For example, the Tower '546 Accused Process includes forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate.

EXHIBIT 36

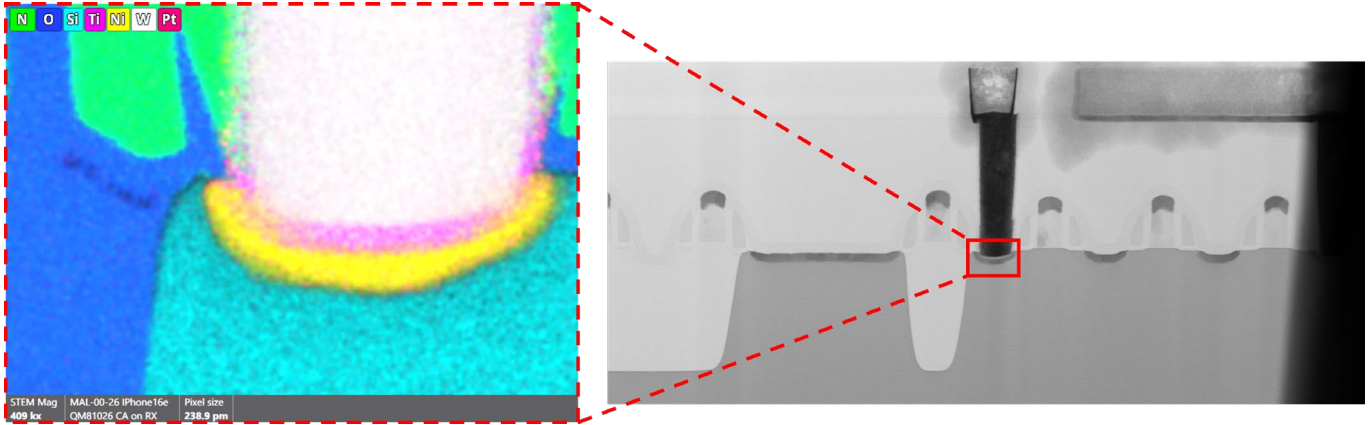
Claim	Tower '546 Accused Process
semiconductor substrate;	 The figure consists of two parts. On the left is an EDS map showing a cross-section of a device with various colored regions representing different elements: N (blue), O (green), Si (red), Ti (yellow), Ni (cyan), W (magenta), and Pt (white). A red dashed box highlights a specific area. On the right is a grayscale SEM image of the same device, with a red square highlighting a contact opening. A red dashed line connects the highlighted area in the SEM image to the EDS map. At the bottom left of the EDS map, text reads: 'STEM Map 400 kv MAL-00-26 iPhone16e QM81026 CA on RX Pixel size 230.9 nm'.
[1b] depositing a first liner layer to conformally coat the contact opening;	The Tower '546 Accused Process comprises depositing a first liner layer to conformally coat the contact opening. For example, the method for manufacturing Qorvo QM81026 chip includes depositing a first liner layer to conformally coat the contact opening.

EXHIBIT 36

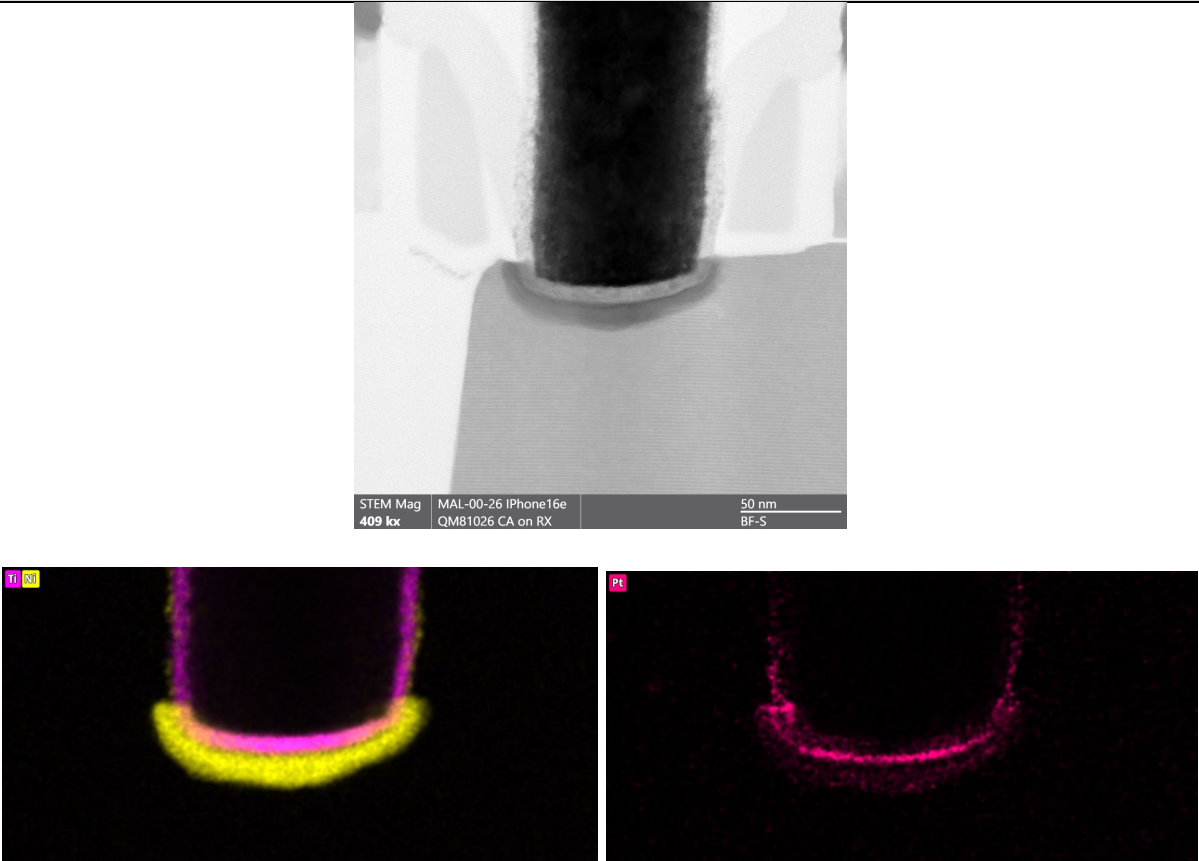
Claim	Tower '546 Accused Process
	 The figure consists of three images. The top image is a grayscale STEM micrograph showing a cross-section of a semiconductor device. A dark, vertical cylindrical feature is centered, sitting on a lighter, trapezoidal base. The background is light gray. At the bottom of the image, there is a black bar with white text: 'STEM Mag 409 lx', 'MAL-00-26 iPhone16e QM81026 CA on RX', and '50 nm BF-S'. Below the STEM image are two smaller, side-by-side images. The left one is an elemental map showing a bright yellow/orange arc at the base of the cylinder, with a small 'Ti' label in the top left corner. The right one is another elemental map showing a bright red/pink arc at the same location, with a small 'Pt' label in the top left corner.
[1c] causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the	The Tower '546 Accused Process comprises causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate. For example, the Tower '546 Accused Process includes causing a portion of the first liner layer to diffuse into the substrate, thereby forming a first intermix region.

EXHIBIT 36

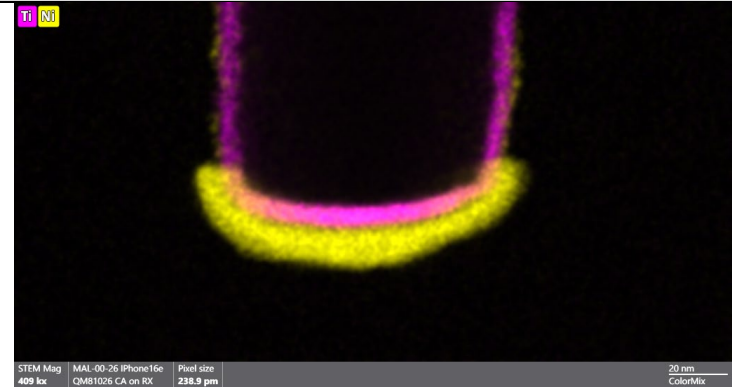
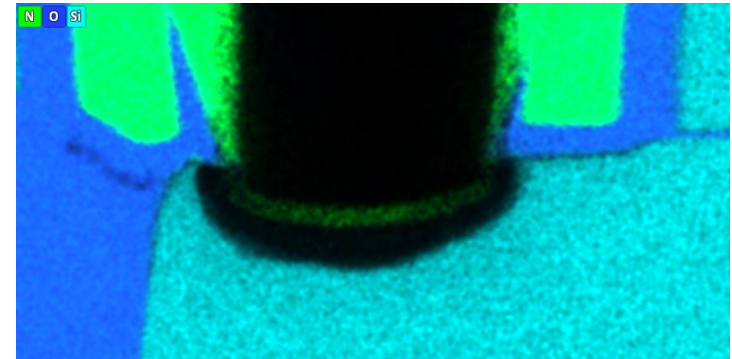
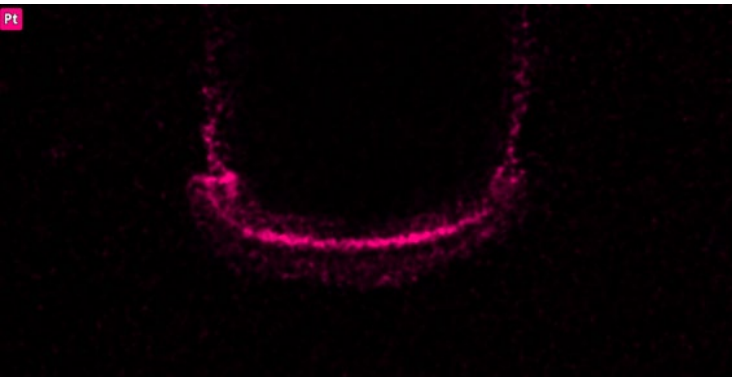
Claim	Tower '546 Accused Process
semiconductor substrate;	 EDS map showing Ti (magenta) and Ni (yellow) distribution. The Ti signal is concentrated in the upper part of the U-shaped structure, while the Ni signal is concentrated in the lower part. The background is black. STEM Mag 409 kx MAL-00-26 iPhone16e QM81026 CA on RX Pixel size 238.9 pm 20 nm ColorMix  EDS map showing N (green), O (blue), and Si (cyan) distribution. The N and O signals are concentrated in the upper part of the U-shaped structure, while the Si signal is concentrated in the lower part. The background is black.  EDS map showing Pt (magenta) distribution. The Pt signal is concentrated in the lower part of the U-shaped structure. The background is black.

EXHIBIT 36

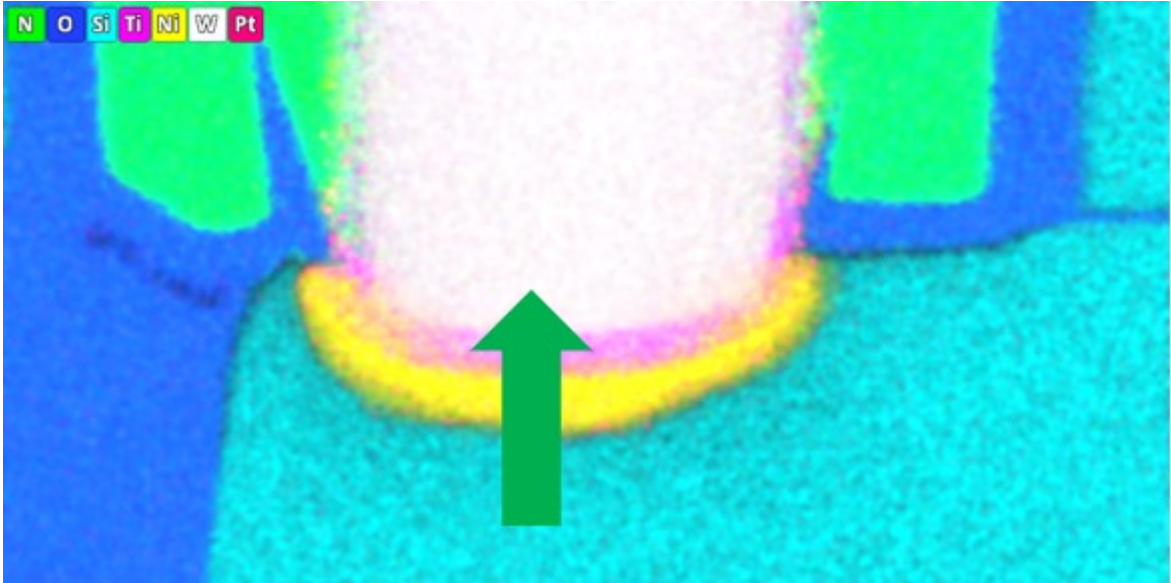
Claim	Tower '546 Accused Process
	The arrow in the reverse engineering analysis below corresponds to the arrow at the top of the ensuing material analysis chart (i.e., such that the upward direction in the reverse engineering analysis below corresponds to the right-to-left direction in the ensuing material analysis chart).  As shown in the material analysis chart below, the intermix region created by the Tower '546 Accused Process includes a portion of the first liner layer diffusing into the silicon substrate.

EXHIBIT 36

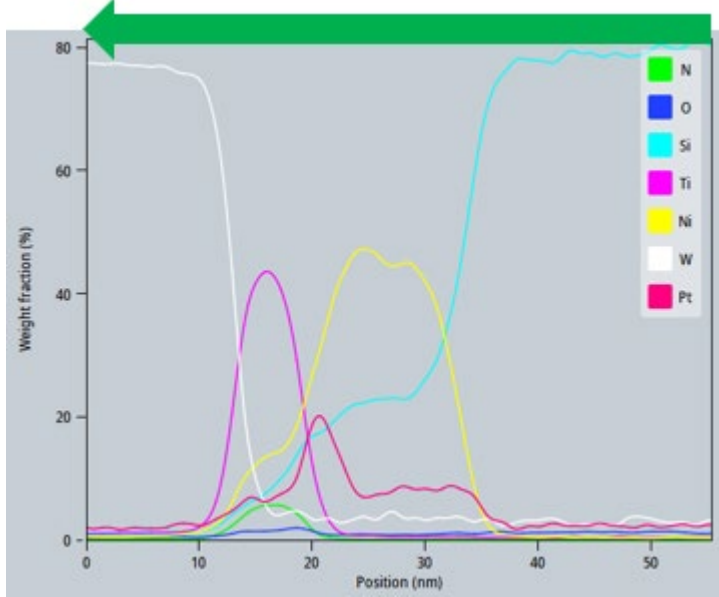
Claim	Tower '546 Accused Process
	 The graph displays the weight fraction (%) of various elements as a function of position (nm) from 0 to 50 nm. The elements tracked are N (green), O (blue), Si (cyan), Ti (magenta), Ni (yellow), W (white), and Pt (pink). A large green arrow points from the right side of the graph towards the left, indicating a reverse engineering analysis. The Si profile starts at ~78% at 0 nm and drops to near 0% by 10 nm. The Ti profile peaks at ~45% around 18 nm. The Ni profile peaks at ~48% around 25 nm. The Pt profile peaks at ~20% around 20 nm. The W profile is near 0% throughout. The N and O profiles are near 0% throughout.
[1d] depositing a refractory metal layer over the first intermix region; and	The Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region. For example, Tower '546 Accused Process includes depositing a refractory metal layer over the first intermix region, as shown in the reverse engineering analysis below.

EXHIBIT 36

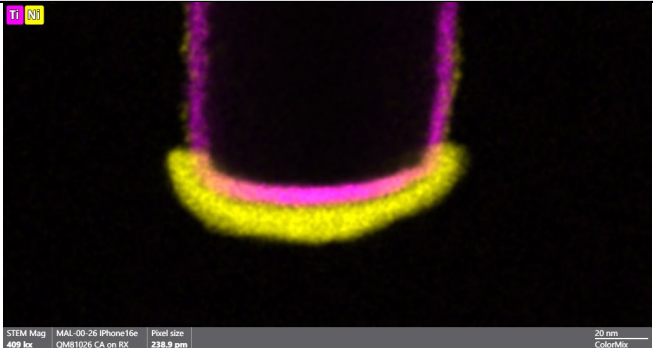
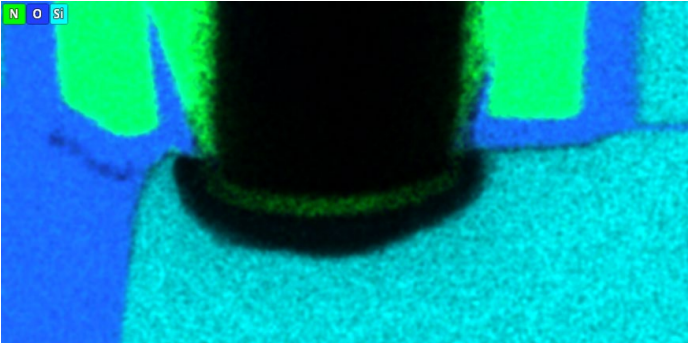
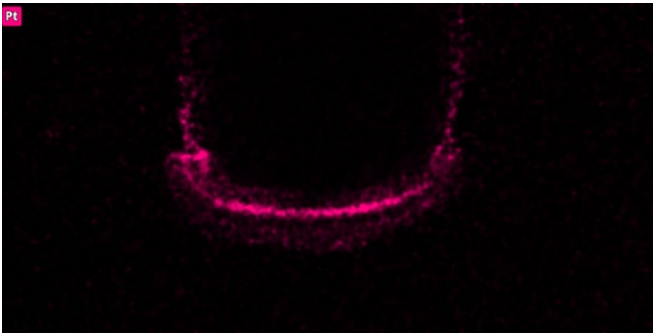
Claim	Tower '546 Accused Process
	 EDS map showing Ti (magenta) and Ni (yellow) distribution. The structure is U-shaped with a thick yellow band at the base. Technical data: STEM Mag 400 kv, MAL-60-75 iPhone16e, OM8100% CA on 32, Pixel size 228.9 nm, 20 nm scale bar, C600A16x.  EDS map showing N (green), O (blue), and Si (cyan) distribution. The structure is U-shaped with a thick cyan band at the base. Technical data: N O Si.  EDS map showing Pt (magenta) distribution. The structure is U-shaped with a thick magenta band at the base. Technical data: Pt.

EXHIBIT 36

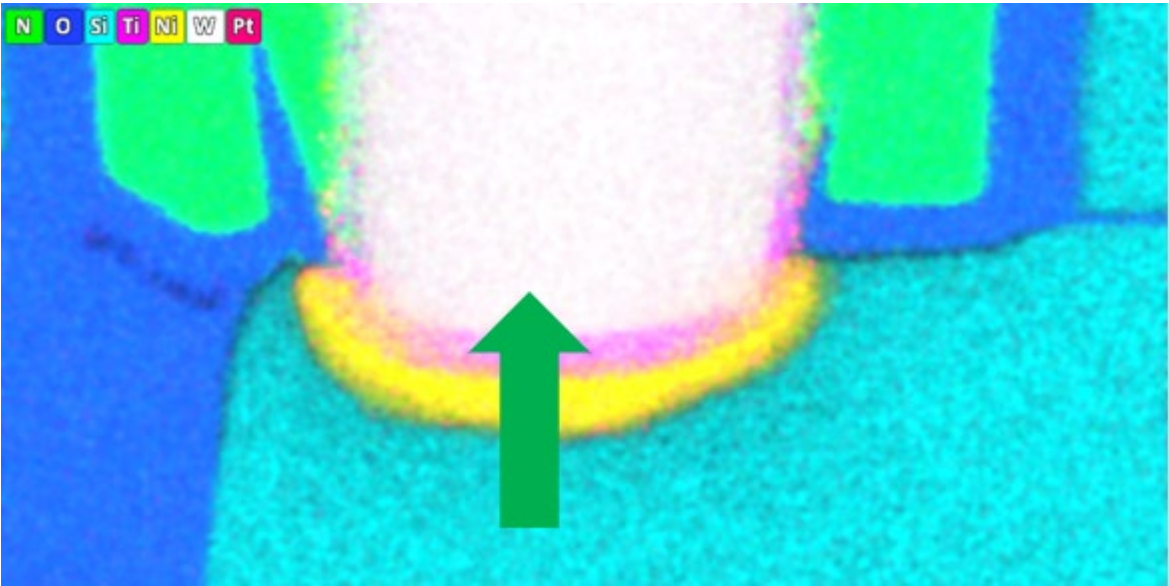
Claim	Tower '546 Accused Process
	Again, the arrow in the reverse engineering analysis below corresponds to the arrow at the top of the ensuing material analysis chart.  As shown in the material analysis chart below, the refractory metal layer is deposited over the first intermix region.

EXHIBIT 36

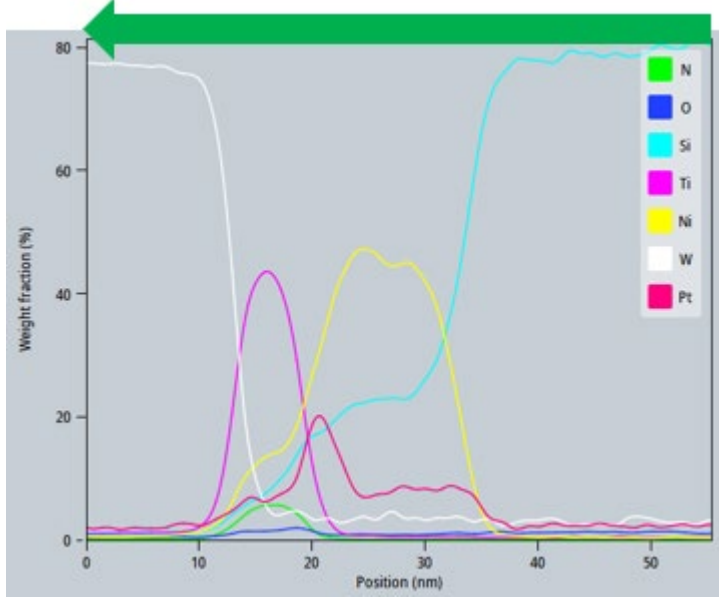
Claim	Tower '546 Accused Process
	 The graph displays the weight fraction (%) of various elements as a function of position (nm) from 0 to 50 nm. The elements tracked are N (green), O (blue), Si (cyan), Ti (magenta), Ni (yellow), W (white), and Pt (pink). A large green arrow above the graph points from right to left, indicating the direction of the analysis. The Si signal (cyan) starts at approximately 78% at 0 nm and drops to near 0% by 10 nm. The W signal (white) starts at approximately 78% at 0 nm and drops to near 0% by 10 nm. The Ti signal (magenta) peaks at approximately 45% around 18 nm. The Ni signal (yellow) peaks at approximately 48% around 25 nm. The Pt signal (pink) peaks at approximately 20% around 20 nm. The Si signal (cyan) rises sharply after 30 nm, reaching approximately 78% by 40 nm and remaining high until 50 nm. The N, O, and Pt signals remain at low levels throughout the scan.
[1e] depositing a metal in the contact opening thereby forming the contact.	The Tower '546 Accused Process comprises depositing a metal in the contact opening thereby forming the contact. For example, the Tower '546 Accused Process includes depositing a metal in the contact opening, as shown in the reverse engineering analysis below.

EXHIBIT 36

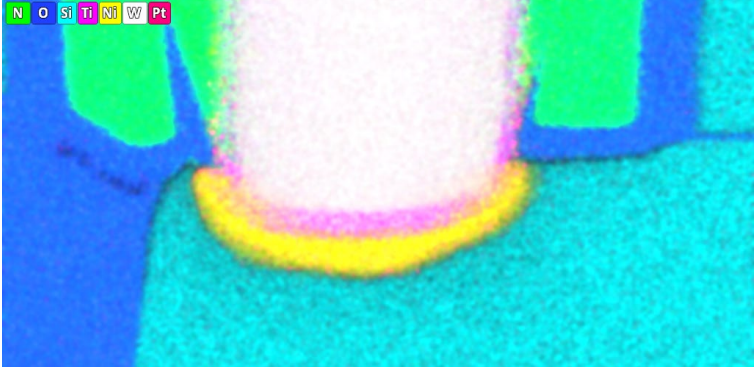
Claim	Tower '546 Accused Process
	 An EDS spectrum image showing a cross-section of a transistor structure. The image displays various elements: Nitrogen (N) in green, Oxygen (O) in blue, Silicon (Si) in light blue, Titanium (Ti) in yellow, Nickel (Ni) in orange, and Platinum (Pt) in red. The structure features a central gate region (pinkish-white) flanked by source and drain regions (cyan), with a yellow/orange layer at the base of the gate.
[8pre] A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:	To the extent the preamble is limiting, the Tower '546 Accused Process comprises a method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain. <i>See, e.g.,</i> limitation [1pre]. The process used to create the source and drain in the Tower '546 Accused Process is the same. The transistors formed by the Tower '546 Accused Process include a gate between the source and the drain.

EXHIBIT 36

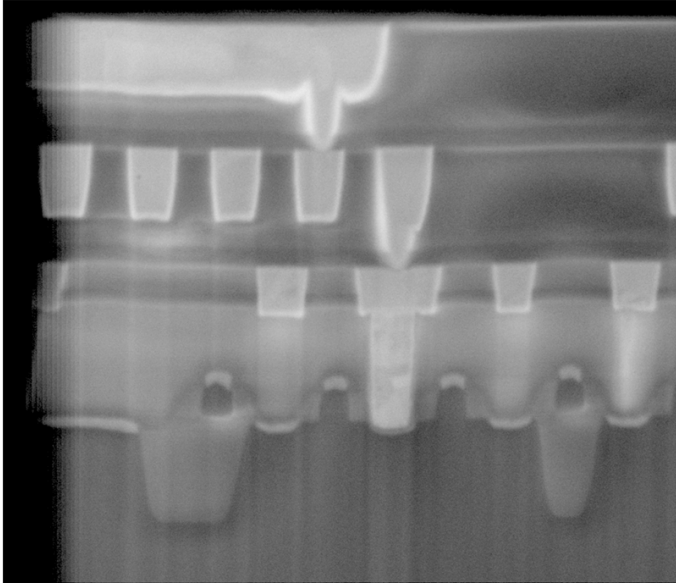
Claim	Tower '546 Accused Process
	
[8a] forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain;	The Tower '546 Accused Process comprises forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain. <i>See, e.g.</i>, limitation [1a].
[8b] depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix	The Tower '546 Accused Process comprises depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain. <i>See, e.g.</i>, limitation [1b] and limitation [1c].

EXHIBIT 36

Claim	Tower '546 Accused Process
region at the upper portion of the each of the source and the drain;	
[8c] depositing a refractory metal layer over the first intermix region; and	The Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region. <i>See, e.g.</i> , limitation [1d].
[8d] depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.	The Tower '546 Accused Process comprises depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts. <i>See, e.g.</i> , limitation [1e].